Survival-guided matrix factorization reveals prognostic transcriptional programs in pancreatic cancer

Amber M. Young[a,1,2], **Alisa Yurovsky**[b], **Xianlu Laura Peng**[c,d], **Didong Li**[a], **Jen Jen Yeh**[c,d,e], **and Naim U. Rashid**[a,d]

[a]Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA; [b]Department of Biomedical Informatics, Stony Brook University, Stony Brook, NY 11794, USA; [c]Department of Pharmacology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA; [d]Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA; [e]Department of Surgery, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

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The transcriptional programs that explain the most variance in bulk tumor expression data are often not those that predict patient survival. Standard nonnegative matrix factorization (NMF) discovers latent structure by minimizing reconstruction error, which may be dominated by variance-dominant but prognostically neutral signals such as tissue composition. Here we present DeSurv, a survival-supervised deconvolution framework that jointly integrates NMF with Cox proportional hazards modeling, shaping gene programs to capture outcome-relevant structure during discovery rather than only in post hoc evaluation. In simulations with known latent structure, DeSurv more reliably recovers the true factorization rank and the identity of prognostic gene programs than standard NMF, with the largest advantage when variance and prognosis diverge. In pancreatic ductal adenocarcinoma (PDAC), DeSurv recovers a compact, three-factor decomposition in which a single tumor–stroma coupling drives prognosis while exocrine-compositional signals are suppressed. These survival-aligned factors generalize across independent PDAC cohorts with consistent hazard ratios (pooled validation HR per unit risk score = 1.78, 95% CI 1.48–2.13, P < 0.001) and clearer survival separation than their unsupervised counterparts, indicating that outcome supervision during factorization yields more transportable transcriptional signatures than the standard discover-then-evaluate paradigm.

nonnegative matrix factorization | survival analysis | semi-supervised learning | tumor deconvolution | pancreatic cancer

Identifying the transcriptional programs that predict clinical outcomes is central to molecular subtyping and prognostic stratification in cancer (1). Bulk transcriptomic profiling remains the dominant technology for the large, clinically annotated cohorts required for such survival modeling, but each expression profile is a composite of malignant cells, cancer-associated fibroblasts, immune infiltrates, and other microenvironmental components (2). As a result, these signatures may be diluted or obscured by variation in cellular composition across samples. Nonnegative matrix factorization (NMF) has been instrumental in resolving this mixture into additive, interpretable gene programs corresponding to recognizable cell types or transcriptional states (3–5). In pancreatic ductal adenocarcinoma (PDAC), this approach has yielded key biological insights, including the identification of molecular subtypes (1), the basal-like and classical tumor programs defined through virtual microdissection (4), and compartment-specific deconvolution of tumor and microenvironmental signals (5).

However, the standard approach discovers these programs through unsupervised factorization (6) and only then evaluates their clinical relevance retrospectively (1, 7). Because unsupervised NMF minimizes reconstruction error, its factors tend to be dominated by the highest-variance patterns in the data, which may reflect tissue composition or other outcome-neutral sources rather than prognostic biology (8, 9). In PDAC, tissue-composition signals such as from exocrine cells reflect low tumor purity rather than cancer cell-intrinsic programs and carry little prognostic value (10, 11). This misalignment between variance and prognosis extends beyond PDAC: tumor purity alone accounts for a large fraction of expression variation across cancer types (12), yet the objective optimized during unsupervised discovery (reconstruction error) differs fundamentally from the criterion used during evaluation (survival association) (9). Identifying the prognostically relevant programs among the discovered factors requires extensive downstream evaluation: over a decade of PDAC subtyping was needed to converge on a robust basal/classical dichotomy, first defined through virtual microdissection (4) and subsequently confirmed in independent cohorts (10), single-cell analyses (13, 14), and prospective clinical profiling (15).

Sufficient dimension reduction theory establishes that response-guided subspace estimation targets directions most relevant to the outcome, whereas variance-maximizing projections can miss outcome-relevant structure entirely (9, 16),

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Significance Statement

Tumor transcriptomes mix malignant and microenvironmental signals, making it difficult to identify programs that drive clinical outcomes. Existing deconvolution and matrix factorization methods discover latent programs but do not ensure prognostic relevance, while supervised predictors often sacrifice biological interpretability. We present DeSurv, a survival-supervised deconvolution framework that integrates nonnegative matrix factorization with Cox modeling to learn gene programs and their survival associations jointly. By embedding outcome information into discovery and using automatic model selection, DeSurv yields clinically relevant, reproducible programs across cohorts. This provides a general framework for identifying prognostically informative gene programs without requiring post hoc filtering or pre-specified signatures, with potential applications wherever high-dimensional nonnegative measurements contain both signal and nuisance variation.

A.M.Y. and N.U.R. conceived and designed the study; A.M.Y. developed the DeSurv method, implemented the software, performed all analyses, and generated figures; A.Y. and D.L. contributed to methodological development and analytic tools; X.L.P. and J.J.Y. contributed to biological interpretation and PDAC domain expertise; A.M.Y. and N.U.R. wrote the manuscript; all authors reviewed and approved the final manuscript.

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[2] To whom correspondence should be addressed. E-mail: ayoung31@live.unc.edu

and outcome-weighted methods have demonstrated this advantage in clustering and principal component settings (8, 17). Whether this advantage extends to NMF-based deconvolution, where nonnegative constraints restrict the factor space geometry, censored survival outcomes provide a noisier supervisory signal, and overly aggressive survival gradients could compromise the mixture-coefficient interpretation of sample loadings, remains an open empirical question.

Here we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox proportional hazards modeling. The key architectural choice is that the survival gradient acts on the gene program matrix W but not on the sample loadings H, directing gene programs toward outcome-relevant structure while preserving the interpretation of loadings as mixture coefficients (3, 18). Because W defines shared transcriptomic programs, new samples can be scored by projection ($Z_{\text{new}} = \tilde{W}^\top X_{\text{new}}$) without requiring their survival data, a property not shared by methods that route supervision through H (19, 20). A supervision parameter α and the factorization rank k are jointly selected via cross-validated concordance (Methods).

We evaluate DeSurv in two settings. In simulations with known latent structure and survival associations, DeSurv recovers the true factorization rank and prognostic programs more reliably than standard NMF. Critically, its advantage scales with the gap between reconstruction-optimal and survival-optimal subspaces and vanishes under null conditions where no survival signal exists, confirming that the method adapts to the structure of the data rather than imposing spurious survival associations. In PDAC, survival supervision produces a different factor structure: DeSurv concentrates survival signal into a single factor coupling classical tumor identity with iCAF-associated stroma, a prognostic axis that unsupervised methods distribute across multiple factors, while the exocrine-compositional variation that dominates standard NMF is not represented as a dedicated factor. These survival-aligned factors generalize across independent PDAC cohorts with consistent hazard ratios and clearer survival separation than their unsupervised counterparts, without retraining or cohort-specific tuning. Because the divergence between variance and prognosis is not unique to PDAC (tumor purity variation dominates expression landscapes across cancer types), DeSurv provides a general framework for extracting outcome-relevant programs from bulk tumor transcriptomes wherever unsupervised factors conflate biological signal with compositional noise. The method is implemented in an open-source R package (https://github.com/ayoung31/DeSurv).

Results

Model Overview. We developed DeSurv, a survival-supervised deconvolution framework (Fig. 1; Methods), and evaluated it in controlled simulations and PDAC cohort analysis with external validation.

Survival supervision clarifies NMF rank selection. We first asked whether incorporating survival outcomes resolves the instability of unsupervised rank selection heuristics (6, 21). Using gene expression data from the PDAC training cohorts (TCGA (22) and CPTAC (23); Methods), we evaluated commonly used unsupervised criteria across a range of candidate ranks.

Standard NMF diagnostics yielded inconsistent guidance (Fig. 2A-C). Reconstruction residuals decreased smoothly with increasing k and did not exhibit a clear elbow, a pattern consistent with both relatively small solutions ($k \approx$ 3-4) and substantially larger ranks ($k \approx$ 6-8). The cophenetic correlation coefficient began to decline at low ranks ($k \approx$ 3-4) but continued to fluctuate at higher values without a distinct transition point. In contrast, mean silhouette width was highest at very small ranks ($k \approx$ 2-3) and decreased monotonically thereafter, favoring low-dimensional solutions that conflicted with the other criteria. Together, these unsupervised heuristics pointed to incompatible values of k, illustrating the ambiguity of rank selection in standard NMF.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance: cross-validated concordance index (C-index), optimized jointly with supervision strength α. The resulting C-index surface identified a coherent region of high performance (Fig. 2D). Model selection followed the one-standard-error rule (Methods): we selected the smallest k whose predicted performance lay within one standard error of the maximum, yielding a parsimonious choice. Applying the 1-SE rule to the Bayesian optimization surface selected $k = 3$ and $\alpha = 0.334$ (peak C-index 0.655; $n = 273$ patients, 139 events). Survival-supervised model selection thus resolves the rank ambiguity that unsupervised heuristics leave open, yielding a single parsimonious factorization for downstream evaluation.

DeSurv recovers prognostic gene programs when variance and prognosis diverge. To test whether survival supervision improves gene program recovery under nonnegative constraints, we designed simulations with known ground-truth latent structure. Simulated expression matrices were generated from a nonnegative factor model in which prognostic gene programs explained low variance relative to outcome-neutral background signals, with survival times generated from latent factor scores via a Cox model ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Methods). To test whether survival information should enter during factorization rather than only afterward, we compared DeSurv to standard NMF ($\alpha = 0$), where factors are learned by reconstruction error alone and survival associations are estimated by a separately fit Cox model. Both methods were tuned via Bayesian optimization over cross-validated C-index, so that differences in performance reflect the role of supervision during factorization, not during model selection.

In the primary scenario, where prognostic programs explained low variance relative to outcome-neutral programs, DeSurv consistently achieved higher C-index (Fig. 3A) and substantially higher precision for recovering the true prognostic gene programs (Fig. 3B), where precision denotes the fraction of top-weighted genes in a learned factor that overlap with a true prognostic program's gene set. The unsupervised baseline exhibited near-zero precision, indicating that variance-driven factorization failed to concentrate on the genes that actually drove survival. This confirms that outcome supervision during factorization is necessary to recover prognostic programs that do not dominate expression variance. Across 100 replicates, DeSurv achieved median C-index 0.839 versus 0.724 for standard NMF ($\Delta = 0.115$, paired Wilcoxon $P = 4 \times 10^{-18}$).

To verify that these gains reflect genuine signal recovery rather than overfitting, we repeated the analysis under a null scenario ($\beta = 0$) and a mixed scenario with partial overlap

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between variance-dominant and prognostic structure (SI Appendix, Fig. S2). DeSurv's benefit scaled with the degree of divergence between variance and prognosis: largest in the primary scenario ($\Delta = 0.115$), attenuated when variance and prognosis partially overlapped ($\Delta = 0.065$, paired Wilcoxon $P = 4 \times 10^{-18}$), and absent when no survival signal existed, where both methods yielded C-index near 0.5. DeSurv also recovered the true rank ($k = 3$) more frequently than standard NMF, which systematically under-selected near $k = 2$ (Fig. 2E).

Survival supervision and standard NMF produce different factor structures in PDAC. Bayesian optimization selected $k = 3$ and $\alpha = 0.334$ for DeSurv in the PDAC training cohorts (TCGA and CPTAC). To compare the factor structures produced with and without survival supervision at the same dimensionality, we fit standard NMF at DeSurv's BO-selected rank ($k = 3$) and examined the overlap between each method's factor-specific gene rankings and established PDAC gene programs (Fig. 4A–B). Matching ranks ensures that differences in factor content reflect the presence or absence of survival supervision rather than differences in model complexity; whether NMF's performance improves at higher ranks is evaluated in the sensitivity analysis below and in SI Appendix, Table S4. We first describe the biological identity of each method's factors, then ask whether these structural differences correspond to differences in survival contribution.

We refer to the three DeSurv factors as D1–D3 and the three NMF factors as N1–N3. In the DeSurv factorization, D1 correlated with both classical tumor programs and an iCAF-associated stromal signature, D2 with activated immune-stromal programs, and D3 with basal-like tumor programs (Fig. 4A). The co-occurrence of classical and iCAF signatures within a single factor is consistent with evidence that survival stratification in PDAC improves when tumor-intrinsic and CAF subtypes are considered jointly (11, 24). No DeSurv factor showed a significant positive correlation with exocrine-associated gene programs (DECODER Exocrine, Collisson Exocrine, Bailey ADEX; Fig. 4A). The survival contributions of these factors and their relationship to expression variance are quantified below (SI Appendix, Fig. S7).

Standard NMF produced a different factor organization (Fig. 4B). One factor (N2) was strongly and significantly correlated with exocrine-associated gene programs and negatively correlated with immune and stromal signatures (Fig. 4B). A second factor (N1) correlated with classical tumor identity (Fig. 4B), while a third (N3) correlated with immune, fibroblast, and extracellular matrix-related programs (including Puleo Desmoplastic, SCISSORS panCAF and myCAF, Maurer Immune and ECM, and DECODER Activated Stroma), aggregating these into a single factor rather than separating them as DeSurv does (D1 and D2; Fig. 4A). At $k = 3$, NMF thus devotes one of its three factors to exocrine-associated variation, leaving two factors to capture all tumor-intrinsic and microenvironmental programs. Whether these structural differences correspond to differences in prognostic information is quantified next.

We quantified per-factor survival contribution using two metrics (SI Appendix, Fig. S7; SI Methods). For each factor, we computed the conditional variance explained (semi-partial R^2: the incremental fraction of total variance accounted for by that factor given all other factors), and its survival contribution, measured as the change in Cox partial log-likelihood ($\Delta\ell$) when that factor is dropped from a k-factor Cox model. All three NMF factors had negligible $\Delta\ell$ despite spanning a range of variance, while DeSurv concentrated nearly all survival contribution into D1. Specifically, D1 explained 7% of variance with $\Delta\ell = 66.4$, while the highest-variance NMF factor explained 23.9% of expression variance but contributed only $\Delta\ell = 0.1$ to survival. The highest-variance DeSurv factor (D3, 35.4% of variance) also contributed minimally to survival after accounting for D1. Thus, in PDAC the highest-variance factors are not the most prognostic, consistent with tumor purity analyses across cancer types (12).

To further characterize how the two factorizations relate, we examined pairwise W-matrix gene loading correlations (SI Appendix, Fig. S8). NMF's classical-associated factor (N1) showed the strongest correspondence with DeSurv's basal-like factor (D3), and NMF's microenvironmental factor (N3) mapped primarily to DeSurv's D2. NMF's exocrine-associated factor (N2) did not correspond strongly to any single DeSurv factor, with its strongest (though weak) correlation to D3, consistent with D3's high variance and low survival contribution. Together, the two methods share overlapping tumor and microenvironmental structure but differ in how exocrine-associated variation is represented: NMF dedicates a factor to it, whereas DeSurv does not. The key quantitative difference is that DeSurv concentrates survival contribution into a single factor ($\Delta\ell = 66.4$), whereas no NMF factor achieves comparable survival contribution at the same rank. Whether this translates to improved generalization across independent cohorts is tested below.

Survival-aligned programs generalize across independent PDAC cohorts. A prognostic factorization is useful only if it generalizes beyond the training cohort. Having established that survival supervision and standard NMF produce different factor structures in the training data, we next tested whether DeSurv's structure transfers to independent cohorts without retraining. We computed factor scores in each validation dataset by projecting the gene programs learned in the training cohort ($Z = \tilde{W}^\top X_{\text{new}}$), yielding a continuous score for each factor per sample.

For DeSurv, we evaluated D1 (the classical–iCAF co-program with the largest training-data $\Delta\ell$); for standard NMF at the same rank, we selected the factor with the largest $\Delta\ell$ (N1, the classical-associated factor). External validation C-index for NMF at independently selected ranks is summarized in SI Appendix, Table S4; standard NMF at elbow-selected $k = 5$ and BO-selected $k = 7$ showed no systematic concordance advantage over NMF at $k = 3$, indicating that additional unsupervised factors do not recover the prognostic signal that supervision concentrates. Across 5 independent PDAC cohorts ($n = 616$), the DeSurv linear predictor showed a consistent survival association (stratified Cox model: pooled HR 1.78; 95% CI 1.48–2.13; $P = 4.4 \times 10^{-10}$; per-cohort forest plot in SI Appendix, Fig. S9; KM curves in Fig. S5C–F). The analogous NMF linear predictor showed a weaker pooled effect (HR NA; 95% CI NA–NA; $P = NA$); a direct comparison using dichotomized risk groups is presented below.

To assess clinical interpretability of these continuous associations, we dichotomized validation samples into high- and low-risk groups using method-specific cutpoints, each independently optimized via cross-validation on the training data.

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The DeSurv linear predictor separated survival trajectories in the pooled validation cohort (HR = 1.93, 95% CI 1.55–2.41, $P < 0.001$; Fig. 4C). The same procedure applied to NMF yielded weaker stratification (HR = 1.48, 95% CI 1.04–2.1, $P = 0.027$; Fig. 4D). Together, these results indicate that incorporating survival information during factorization yields gene programs with more transportable prognostic associations than the standard discover-then-evaluate approach.

Factorization rank K=3 is robust across supervision settings. The 1-SE model selection rule chose $k = 3$ over the global BO maximum ($k = 7$; CV C-index 0.655 versus 0.646; margin 0.009). To assess whether this reflects a robust solution or an artifact of a specific tuning configuration, we conducted a sensitivity analysis across $K \in \{2, 3, 5, 7, 9\}$ and $\alpha \in \{0, 0.25, 0.35, 0.55, 0.75, 0.85, 0.95\}$ (35 combinations), holding all other hyperparameters at their BO-selected values and evaluating each combination in independent external validation cohorts via stratified Cox regression (SI Appendix, Tables S1–S3). $K = 3$ achieved significance ($P < 0.05$) across 6 of 7 supervision strengths, and $K = 7$ at a comparable 5 of 7. After additional adjustment for PurIST and DeCAF molecular classifiers, $K = 3$ retained significance at 5 of 7 supervision strengths, while $K = 7$ dropped to 2 of 7 (SI Appendix, Table S2), indicating that much of $K = 7$'s prognostic signal overlaps with existing molecular classifiers. The parsimonious $K = 3$ solution thus validates robustly across supervision strengths and retains prognostic value beyond known classifiers, providing empirical support for the 1-SE rule.

Discussion

Aligning the discovery objective with the evaluation criterion, incorporating survival outcomes directly into NMF factorization rather than assessing them retrospectively, produces a different and more prognostically transportable factor structure in PDAC. DeSurv concentrates prognostic signal into fewer, more interpretable gene programs that generalize across independent cohorts without retraining. More broadly, the principle that outcome-guided dimensionality reduction targets different subspaces than variance-driven reduction extends beyond cancer genomics: sufficient dimension reduction theory (9) and the information bottleneck framework (25) both predict that supervised compression retains outcome-relevant structure while discarding nuisance variation. DeSurv implements this principle within the specific constraints of NMF deconvolution, where nonnegativity preserves biological interpretability and the factorization structure enables single-sample scoring by projection.

In PDAC, the factor structure produced by survival supervision is biologically informative. The factor with the largest survival contribution (D1) couples classical tumor programs with an iCAF-associated stromal signature, consistent with growing evidence that clinical outcomes depend on the joint configuration of tumor-intrinsic and cancer-associated fibroblast subtypes (11, 24). This co-occurrence within a single factor suggests that the prognostically relevant biology in PDAC is not purely tumor-intrinsic but reflects tumor–stroma coupling that unsupervised methods distribute across multiple factors. Standard NMF, by contrast, allocated a dedicated factor to exocrine-compositional variation, a signal that reflects tumor purity rather than cancer biology (10), leaving less capacity for the microenvironmental programs that DeSurv separates. The methodological contribution lies not in the identity of these programs, which have been established through virtual microdissection (4), experimental microdissection (26), and unsupervised deconvolution (5), but in how they are recovered: de novo, without pre-specified signatures, with direct quantification of their survival associations during factorization.

The semi-supervised design reflects a deliberate balance between biological completeness and clinical relevance. Survival supervision concentrates prognostic signal into fewer factors, creating a broad concordance plateau that the one-standard-error rule then leverages to select parsimonious ranks. In PDAC, the concordance plateau for DeSurv spanned $k = 3$–12 (Fig. 2D), indicating that three supervised factors captured nearly as much prognostic information as twelve; for standard NMF, concordance generally increased with k (SI Appendix, Table S4), consistent with the need for additional unsupervised factors to capture prognostic signal that supervision concentrates. This parsimony is a property of the 1-SE rule rather than an intrinsic consequence of supervision, but supervision creates the flat concordance surface under which the rule can operate effectively.

Our simulations provide further guidance on when this tradeoff is most beneficial: DeSurv's advantage is greatest when prognostic programs explain modest variance relative to outcome-neutral signals and vanishes under null conditions where no survival signal exists. DeSurv is therefore not intended to replace unsupervised NMF in all settings. When the goal is exploratory discovery without clinical endpoints, when survival data is sparse or unreliable, or when the application requires comprehensive biological characterization rather than prognostic stratification, unsupervised methods remain appropriate. The α parameter, tuned via Bayesian optimization, lets the data determine the appropriate balance for a given cancer type and cohort size, and the $\alpha = 0$ endpoint recovers standard NMF as a special case.

Several limitations define the scope of these findings. First, DeSurv assumes Cox proportional hazards, which may not hold in settings with delayed or non-proportional treatment effects; the convergence guarantee (SI Appendix) applies to an idealized algorithm, and the implementation includes practical modifications that depart from the theoretical analysis, though empirically these do not affect convergence behavior. Extensions to more flexible survival models are a natural direction. Second, DeSurv was applied here only to PDAC, and broader benchmarking across cancer types is needed to establish the generality of the variance–prognosis divergence pattern; the computational cost of Bayesian optimization may limit scalability to very large cohorts, though hyperparameter selection is performed once and the final model applies by projection. Third, because supervision incorporates observed survival, which reflects both tumor biology and treatment received, the learned factors may implicitly encode treatment-response associations present in the training cohort. The biological interpretation of supervised factors should therefore be understood as conditional on the therapeutic context in which they were derived, and transportability to settings with substantially different treatment landscapes requires empirical validation. Extending DeSurv to multi-omics integration, spatial transcriptomics, and alternative outcome models is a natural next step. As single-cell cohorts with clinical an-

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notation grow in size, patient-level survival modeling may complement deconvolution-based approaches, though the cohort sizes required for stable survival analysis remain available primarily in bulk expression data.

More broadly, the standard workflow in cancer transcriptomics, discovering expression programs unsupervised and then evaluating their clinical relevance retrospectively, incurs a structural cost whenever the highest-variance signals are not the most prognostic. Because tumor purity and tissue composition dominate expression variation across cancer types (12), this mismatch is likely the norm rather than the exception. DeSurv provides a principled alternative: by aligning the factorization objective with the clinical question from the outset, it recovers prognostic programs directly, quantifies their survival contributions during discovery, and produces compact signatures that transfer to new patients by projection. The framework is general to any setting where NMF-based deconvolution is applied and clinical outcomes are available.

Materials and methods

A. Problem formulation and notation. Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the censoring indicators ($\delta_i = 1$ if the event is observed), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$ and linear predictor $Z^\top \beta$.

B. The DeSurv Model. DeSurv integrates Nonnegative Matrix Factorization (NMF) with penalized Cox regression to identify gene programs associated with patient survival. The method operates in a semi-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0,1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W,H,\beta) = (1-\alpha)\,\mathcal{L}_{\mathrm{NMF}}(W,H) - \alpha\,\mathcal{L}_{\mathrm{Cox}}(W,\beta), \quad [1]$$

where $\mathcal{L}_{\mathrm{NMF}}(W,H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W,\beta)$ is the elastic-net penalized partial log-likelihood. A critical architectural choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients (3, 18). When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF. The Cox component also accommodates additional sample-level covariates (e.g., tumor stage or grade) alongside Z, enabling adjustment for known prognostic factors during optimization.

Optimization alternates updates for H, W, and β; although non-convex, these updates converge to a stationary point under mild conditions (SI Appendix). Complete derivations and algorithmic details are provided in the SI Appendix.

C. Hyperparameter selection and cross-validation. Hyperparameters $(k, \alpha, \lambda_H, \lambda, \xi)$ were selected by maximizing the cross-validated C-index using Bayesian optimization (BO), with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lay within one standard error of the maximum). All model selection was performed entirely within training data using nested cross-validation; no validation cohort data were used during any stage of tuning. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (SI Appendix). Before validation, each column of W was truncated to its BO-selected number of top genes (details in SI Appendix), denoted $\tilde{W}$. In PDAC, BO selected $n_{\mathrm{top}} = 270$ top genes per factor. A $k \times \alpha$ sensitivity analysis confirmed the robustness of the selected rank across supervision strengths (see Results and SI Appendix, Tables S1–S3).

External validation was performed by projecting new datasets onto the learned programs via $Z = \tilde{W}^\top X_{\mathrm{new}}$ and evaluating survival associations using C-index and log-rank statistics. Because the learned W is fixed at training time and validation samples are scored by simple projection, no retraining or access to validation survival data is required. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. To evaluate the quality of DeSurv-derived gene signatures for subtyping, new datasets X_{new} were clustered on genes in $\tilde{W}$, and survival differences were analyzed for the derived clusters. Further training, validation, and runtime details appear in the SI Appendix.

D. Simulation studies. Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\mathrm{Gamma}(\mathrm{shape} = 3, \mathrm{rate} = 0.8)$ and cross-loadings onto other factors from $\mathrm{Gamma}(1, 20)$; background gene loadings from $\mathrm{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\mathrm{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\mathrm{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the primary scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T \tilde{W}$, where $\tilde{W}$ retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribu-

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tion. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a primary scenario where prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

E. Real-world datasets. We analyzed publicly available RNA-seq and microarray cohorts of pancreatic ductal adenocarcinoma (PDAC) with corresponding overall survival outcomes. Gene expression matrices were converted to TPM, log-transformed, and filtered to remove low-expression genes. Survival times and censoring indicators were taken from the associated clinical annotations. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and five were used for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo). To harmonize differences in scale across cohorts, filtered gene expression data were within-subject rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility). More details about the datasets can be found in the SI Appendix.

F. Data, Materials, and Software Availability. All datasets used in this study are publicly available and de-identified; no IRB approval was required. Training data were obtained from the Genomic Data Commons (TCGA-PAAD and CPTAC-3). Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830; Puleo, E-MTAB-6134), GEO (Moffitt, GSE71729), and the ICGC Data Portal (PACA-AU array and RNA-seq). Cohort details, sample sizes after quality control, and patient-level metadata used are summarized in SI Methods. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study (11). An R package implementing DeSurv is available at github.com/ayoung31/DeSurv. Code and processed data to reproduce all analyses are available at github.com/ayoung31/DeSurv-paper.

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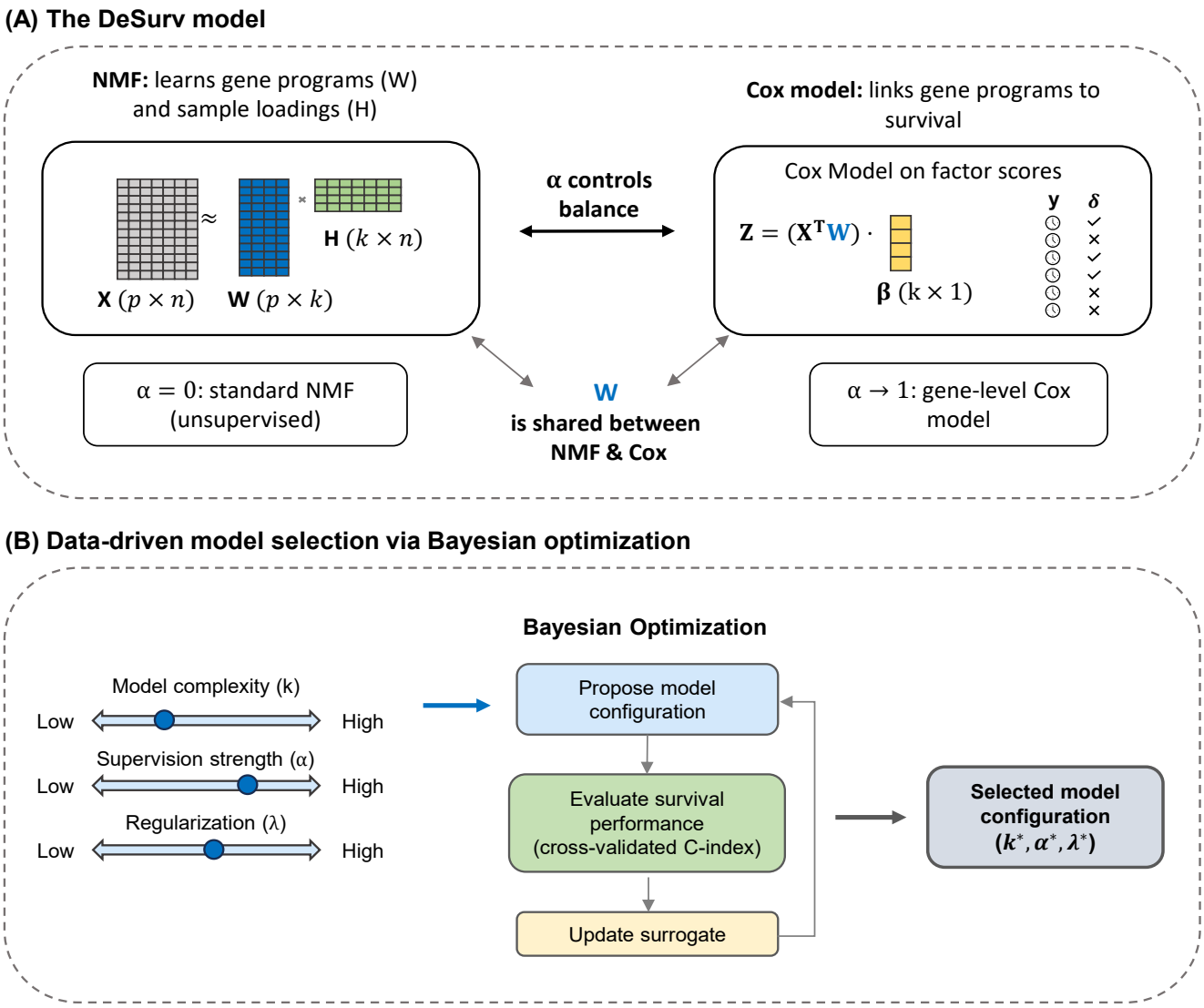

Fig. 1. Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

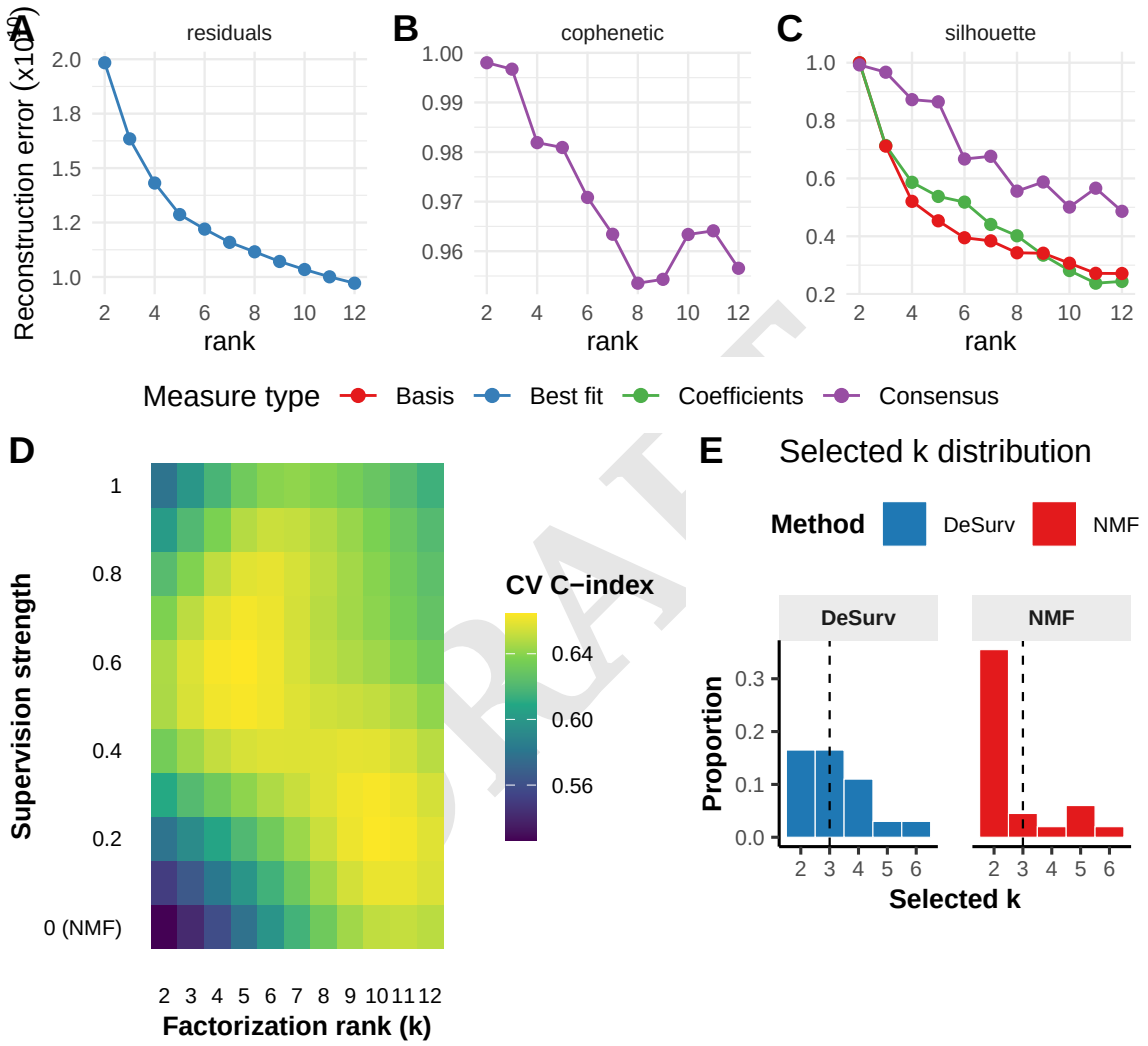

Fig. 2. (A–D) Analyses based on pancreatic ductal adenocarcinoma (PDAC) gene expression data from TCGA and CPTAC cohorts. (A–C) Standard unsupervised rank selection heuristics yield inconsistent guidance for selecting the factorization rank k. (A) Reconstruction residuals as a function of k. (B) Cophenetic correlation coefficient as a function of k. (C) Mean silhouette width across multiple distance metrics as a function of k. (D) Gaussian process predicted mean cross-validated concordance index (C-index) from Bayesian optimization over the joint space of factorization rank (k) and supervision strength (α). (E) Simulation studies with known rank ($k = 3$): distribution of selected k values across repeated replicates for DeSurv versus standard NMF with post hoc Cox modeling ($\alpha = 0$).

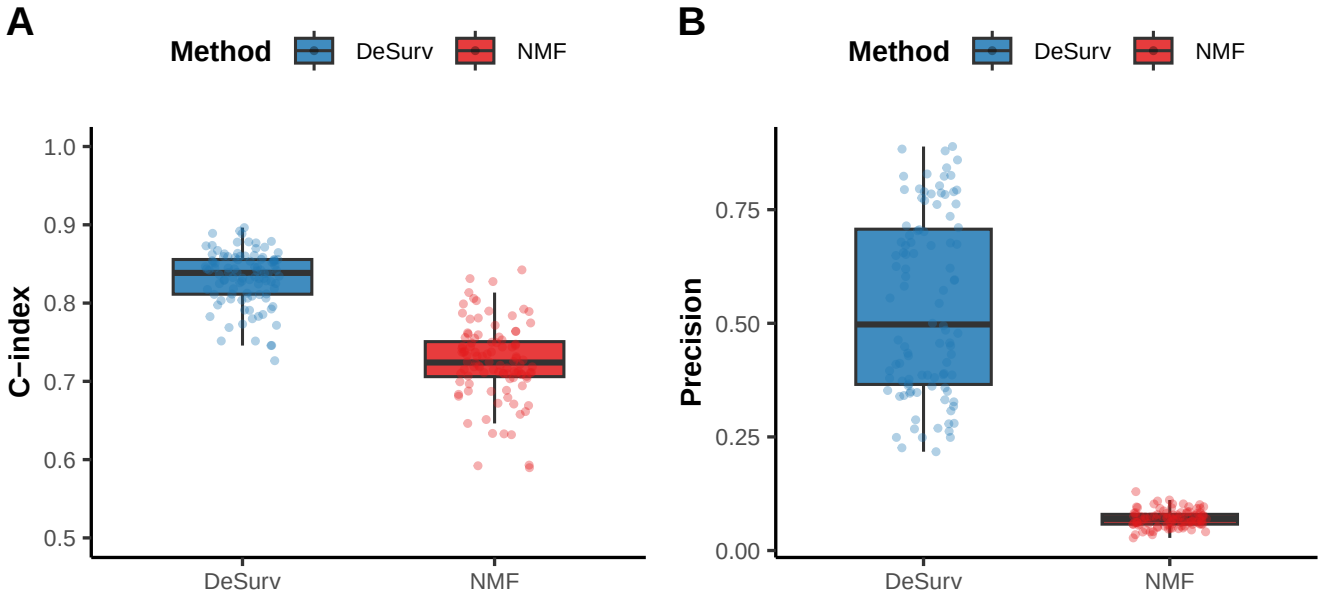

Fig. 3. Performance comparison between DeSurv and standard NMF ($\alpha = 0$) in the primary simulation scenario ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$, 100 replicates), where prognostic programs explain low variance relative to outcome-neutral programs. Both methods were tuned via Bayesian optimization over cross-validated concordance index. (A) Test-set concordance index across simulation replicates. (B) Precision (fraction of top-weighted genes in a learned factor that overlap with a true prognostic program's gene set) across simulation replicates. Each point represents one replicate.

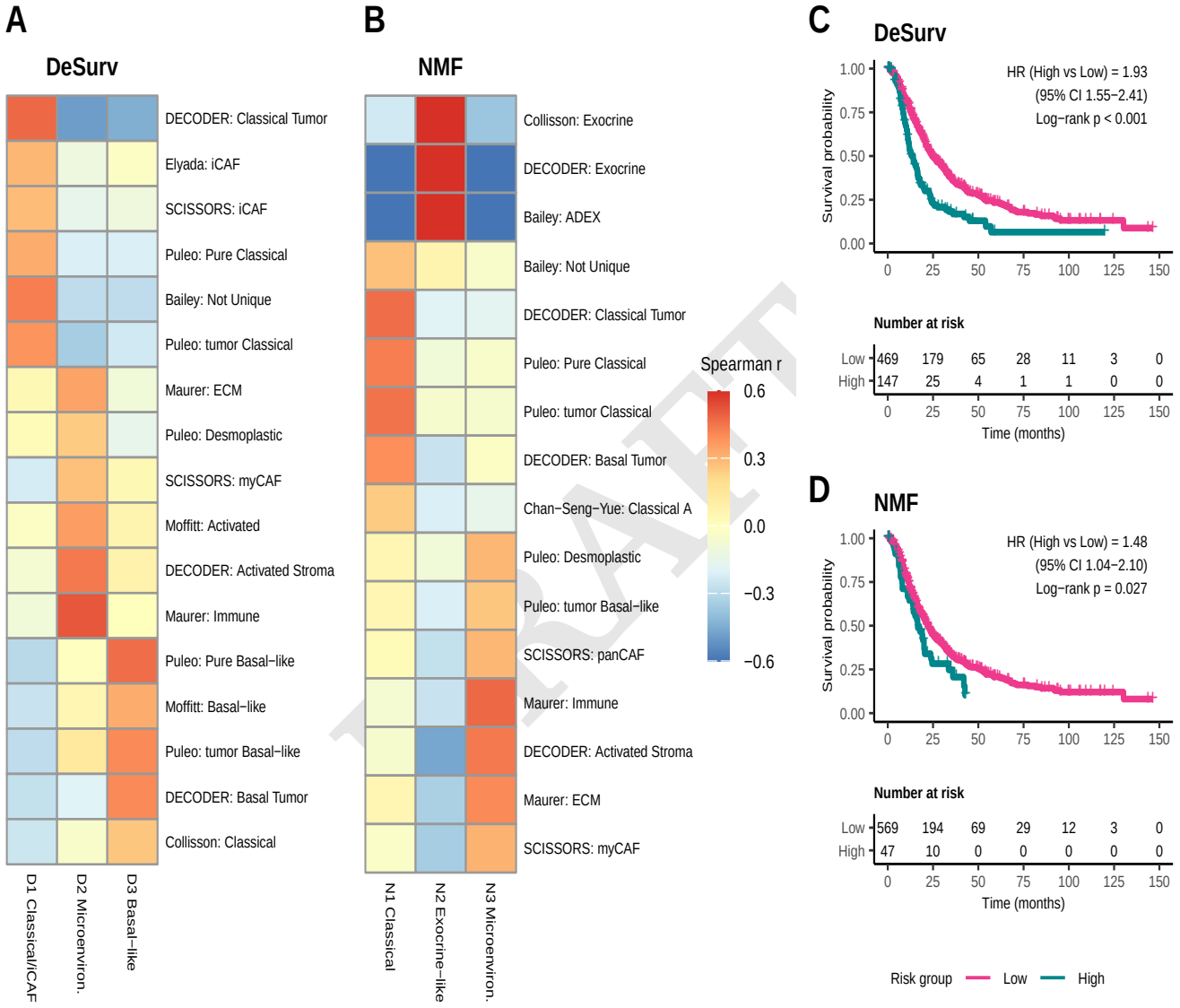

Fig. 4. Survival supervision produces different factor structures that generalize to held-out cohorts. (A–B) Spearman correlations between factor gene rankings and established PDAC gene programs for DeSurv (A) and standard NMF (B). Asterisks denote significance after multiple testing correction; only correlations > 0.2 shown. (C–D) Kaplan–Meier curves for pooled external validation samples stratified by the cross-validated optimal cutpoint for the DeSurv linear predictor (C) and the analogous NMF-derived factor (D).